

èfjç» - ç” µè`❖i¼š188-5665-3017 - é,❖❖±i¼šhs3434@foxmail.com -
ä, ººç½ ‘ç«™i¼šhs3434.github.io - **æ±
,è❖Ææ,,❖â❖ “**i¼šAMD ROCm Python â❖Žç«`â¼â❖ ‘â • ¥ç`â, ^i¼^ROCm Radeon Cloud
æ-¹â❖ ‘i¼% --- ## ä, ººç❖«ä» 3 â¹´ç” Ýç%©äzjæ❖`â • ¥ç`âÇ-ç»❖éâEi¼Çè ± ºâ|è½-âž`
K8s â❖Žç«`æ ^â€, \*\*âº¹â™`âÇ-éç`ç½²â❖èµ,,æº❖èºfâº|â❖ä»»âŠiç¼-æž’ â❖GPU
æž`ç❖ + ææ❖âŠi**â€, ææèèç 2 â “è ± â| ROCm / K8s / Helm / FastAPI /
Prometheusi¼Çç«`ç«`âºÇæ^❖ [**k8s-llm-runtime**](https://github.com/hs3434/k8s-llm-
runtime)i¼^K8s ä, Šçš,, vLLM æ`jâž`ææ❖âŠiç½ ‘â...³i¼%oi¼Ç**v0.1.0
taggedi¼Çâ❖`çžºâºè • ‘demo**â€, ç + Ýæ,% Snakemake ç±» Argo Workflows çš,, ML
â¹³â❖æš½è ± jâ€, --- ## æ±,è❖Çæšèèf½i¼^æÇ%â²-â½❖âÇ¹é...❖æž’ âº❖i¼% |
ç±»â`« | æšèèf½ | |-----|-----| | \*\*â❖Žç«`æ , âçf** | Python 3.11+â❖FastAPIâ❖pytest | |
**âº¹â™`â, žç¼-æž’ ** |
Dockerâ❖Kubernetesâ❖Helmâ❖é • æâf❖ä» “âº “çºjç❖ + i¼^DaoCloud
é • æâf❖âš éèYi¼% | | **GPU / LLM ç” Ýæ❖❖** | vLLMâ❖AMD
ROCmâ❖PyTorchâ❖Hugging Face | | **âº`è Š, æµ«æ Š ** | Prometheusâ❖structlog | |
â • ¥â... é “¾ | uvâ❖ruffâ❖mypyâ❖GitHub Actions | | **èç❖ç»” ** | Linux èç❖ç»’ | ---
éi¹çºç»»éâÇ ### k8s-llm-runtimei¼šK8s ä, Šçš,, vLLM æ`jâž`ææ❖âŠiç½ ‘â...³i¼^v0.1.0
taggedi¼% **é-ºéç`\*\*i¼^â`¹âº” JD è❖Çè`£ 1 + 3i¼%oi¼švLLM æ`jâž`éç`ç½²éæÇ GPU +
âº¹â™` + è • -ç” ±i¼Çâºé`Ýâšâºâ...±âº«æ-¶ æ`jâž`çºjç❖ + âºæ❖, i¼ç” çº¶ | /
âºâºé`Ýæf³ç” “ OpenAI â...¼âº¹ API æ-¶ é-” æ Š é«~â€, i¼^çº`æž¥â`¹âº”
JD i¼š"âº Š æ`jâž`èç»»f/æž`ç❖ + Workshopâ❖æ • ™ç`â❖âšÝèf½æ¼” çº"i¼%
æž ¶ æž,,i¼^âºÇæ chart + ä, %â±, lib + é • æâf❖â¼^âÇ-i¼%oi¼š - `charts/llm-inference/
`i¼švLLM è`Ýè½½ charti¼Ç**GPU vendor â` ± æ❖çä»...é❖ `values.yaml` **i¼^amd.com/
gpu / nvidia.com/gpu / none ä, %ææ❖i¼Çævalues æ¾¾¾¾éæ%æ«©i¼% - `charts/llm-router/
`i¼šFastAPI ç½ ‘â...³ chart - ConfigMap æ³” â...¥ initContainer âš è½½½â...¾¼Ý chart - RBAC
ææèâºæ❖fé™i¼^Pod/Service/ConfigMap read+write + Leasei¼% - ServiceMonitor âº`éè
%âº`ç” “i¼^Service 0.0.0.0:9090/metricsi¼% - `src/k8s_llm_runtime/`i¼š3 â±, lib æš½è ±
j - `K8sJobOperator`i¼^â½žâ±, i¼Çééšç” “ Job èºfâº|i¼% - `VLLMInferenceOperator`i¼^â, -
â±, i¼ÇHelm CLI subprocess + GPU vendori¼% - `ModelOperator`i¼^é«~â±, i¼ÇOpenAI â...
¼âº¹ + è ± ºâš`éç`ç½²i¼% **4 âº Š æšèææ-âºç, 1**i¼š 1. **lib ä, %â±, â` + â±, æš½è ±
j**i¼^â`¹âº” JD â❖Žç«`æž ¶ æž,,èf½âš i¼% - ä, šâ±, ModelOperator âºâ...³âçf
load_model(alias) / unload_model(alias) - ä, â±, VLLMInferenceOperator âºè£... helm
ç» + èš, - â½žâ±, K8sJobOperator ééšç” “ K8s Job èºfâº| 2. **Helm ConfigMap æ³” â...¥
aliases**i¼^â`¹âº” JD âº¹â™`âÇ- + èµ,,æº❖èºfâº|i¼% - é❖çâ...❖æ`æ-æ-æ-ºâçæ`jâž`éç½
rebuild é • æâf❖i¼^GPU é • æâf❖ 15-20GB rebuild 5-10mini¼% - ConfigMap
ç Š çº Š ç” Ýæ • ^i¼Çé¶ ¶ é • æâf❖é ± âºº 3. **K8s Lease â` + â, fâ¼âºé” ❖\*\*i¼^â`¹âº”
JD é» + ç¾æèç❖ç»’ + HAi¼% - coordination.k8s.io/v1 Lease é²æç Router
âšâº% - ææ-â¹ ¶ â❖ ‘éç`ç½²âºÇæ, ææ`jâž`- é” ❖ç²” âº|æÇ% model
aliasi¼Çè ± ºâš`èç ± ææÝ + ç»çº| 4. \*\*ä, šâŠiçº Š Prometheus metrics\*\*i¼^â`¹âº” JD
â¹³â❖âº`è Š, æµ«æ Š i¼% - `chat_completions_duration_seconds`i¼^histogrami¼Çâ❖«
status code æ ± ç¾i¼% - `http\_requests\_total`i¼^counteri¼ÇæÇ% method/route/statusi¼%
- `models\_loaded`i¼^gaugei¼Çâ❖æ~ in-memory statei¼% - ServiceMonitor âº`éè%èç»
Grafana **GPU vendor â` ± æ❖ç demo\*\*i¼^â`¹âº” JD âš â` + GPU é» + ç¾æi¼%oi¼š -
values.yaml `gpu.vendor=amd|nvidia|none` - AMDi¼šamd.com/gpu èµ,,æº❖èºfâº|i¼ÇROCm
vLLM é • æâf❖ - NVIDIAi¼šnvidia.com/gpu èµ,,æº❖èºfâº|i¼ÇCUDA vLLM é • æâf❖ -
nonei¼šCPU only æ`jâ¼âºi¼^kind cluster é»~èºæi¼Çææ-âºæâ¼â❖ ‘ç” “i¼% **è
`è ± âççè`❖**i¼š - 31 commits / 90 tests pass / coverage 88% - ruff + mypy strict + helm-lint

ä... clean - 2 Helm chartsi¼^llm-inference + llm-routeri¼% **è , ©ä ‘ä , Žäž®ä**i¼š -
Service DNS ä , äCE¹é... bugi¼š --set fullnameOverride={release_name} ` äž®ä - kind
èš,ç,¹ä½ä + ...é • öäfäš ééYi¼šcontainerdConfigPatches + DaoCloud mirror -
èž™äºéf½ä + ™ä...¥ ` docs/amd-interview-demo.md ` ä¾,çžºäœä,èé
æœæ¥i¼^v1.1i¼%oi¼šSSE æµä¼äž`ç + / RBAC æ” ¶ ç´ š äè` » / çœYäž GPU
é + ç¾æe2e ### äYºäž Snakemake çš, ML
ä¹³ä • ¥ä½œæµæj + æž ¶ i¼^ç” YäžjèfCEæ™i¼CEç±» Argo Workflowsi¼% -
èfCEæ™i¼šSnakemake rule è æ³ • æ— æ³ • äžžžæ`jä—ésæ,,ç»,ä^ -
è¾è®i¼šæ “ä¼f rule è æ³ • i¼CEç´ æž¥èºfä + ...éf`æž¥ä£äš`æéç” Yæ^
rulei¼CE è š ,èCEfè¾ “ä...¥è¾ “ä ±ºç±»äžä³äç š š æœ`ä¼äæ¼äž¥ - ä...³è” i¼šJD
äš ä^ + éj¹æœä™`ä|ä¹ ä¹³äç” ä ‘ç»éªCE`â€” â€” æœ-éj¹çºæ™ ML ä¹³äæš½è±
jçš,,ä • ¥ç`äºžè • µ ### è,, ‘æœæž¥ä£ CNN è š £ç ä™i¼^PyTorchii¼%â€” â€” 6 ä “ä -
|ä¹ è • çºžäº š ä ±º - ç«äºç«æ— ¶ çºä • ç š è š £ç EEG äžjä • - PyQt äžæ— ¶ GUI +
FastAPI æš´éœ² - **ä» “äº “**i¼š[github.com/hs3434/bci-pipeline-demo](https://github.com/
hs3434/bci-pipeline-demo) --- ## ä • ¥ä½œç»äž + ### ä , šæµ • äº
æé ± CEäžjæç š ‘æšæœ%œä...-ä , i½œ ç” Yç%äžjæ-ä • ¥ç`ä , ^ i½œ
2025.04 - è ± ºäš - **Docker éf`ç½²**i¼šä , šäšjæ£ç Docker äCE-i¼^äYºäž snakemake +
è ± ºç” æj + æž ¶ i¼% - **æœäšjç«èžç»`**i¼šLinux æœäšjæ™æ—
¥ä , çç´ æšæi¼CEnginx äää “ä»£ç + i¼CEsystemd æœäšjçjç +
äº ‘ä¹³äºä • ¥ä... • ä¼ä “i¼šä , šäšjè ± ºäš`äCE-æµç` + çºæ³ • ä • ¥ä... • ###
ä , šæµ • æ-š æ~ “ç” Yç%äCEä|æœ%œä...-ä , i½œ ç” Yç%äžjæ-äºžä¹ ç” Y i½œ
ç” Yç%äžjæç” ä ‘ä • ¥ç`ä , ^ i½œ 2022.08 - 2024.07 - **äº ‘ä¹³äç” Yäžjä • ¥ä...
• ä¼ä “**i¼šR / Python / Linux / Docker / äº ‘äžYç” Y - **ä , šäšjè ± ºäš`äCE-**i¼šä»ä -
|è½-ä½ • ç»,ä , šäšjæµç`è ± ºäš`äCE-i¼CEæ`jä—äCE-äæä-äµç” -
ä¹³äæäºäi¼š[cloud.oebiotech.com](https://cloud.oebiotech.com/#/home) ###
ä , šæµ • æ-š æ~ “ç” Yç%äCEä|æœ%œä...-ä , i½œ ç” Yç%äžjæ-äºžä¹ ç” Y i½œ
2021.10 - 2022.01 - ä • ç» + èfžè½-ä½ • ç»,i¼šè
`æž š æä , è š ,ä^ + æžæä” ®äžæä , æé š äCE-ä^ + æž -
ä , »è|æšæœ-i¼šä • ç» + èfžè½-ä½ • ç»,æäRæPythonæLinux --- ##
æ • ™è,²èfCEæ™ **è¥žä—ä + œæž-ç š ‘æšä š ä|**i½œ æçç%ç š ‘ä -
|ä , žæšæœ-ä , “ä , š i½œ æœ-ç š ‘i½œ 2018.09 - 2022.07 i½œ GPAi¼š3.35 - æ-ä , šèºæ-
± i¼šäšäYºäžéè¾ ‘æ-è’ ,äžä½ çjºäšäºé|ä€ ä¼ç , ä...³æé š çš ¶ äšæš—
ä€ æé š çš ¶ æCE ± æ ± ä€ - äµ š ä^éj¹çºi¼šæ¹çºææžçç š ä¹ç´ èš±èœè “žæš—
é “ææ±jæY “çš,,ä½±ä “i¼^ä ,ä , žä½œèè...i¼% --- ## ä , äºä¼~äšž - **6 ä “è ± ºä| K8s
äžç«æ`**i¼šäžé¶ ä^º v0.1.0 tagged éj¹çºi¼CEç«äCEæ^äCE Helm chart + Python
lib - **çžºæº demo èf½äš`\*\*i¼šä-è ± ºä , |ç” µè,, ‘äœ` kind cluster ä , šè • ‘éšäæ • ´
LLM æž`ç + æµç`i¼^éf`ç½² ä + è • ç” ± ä + chat completions ä + metricsi¼% -
ä|ä¹ èf½äšä¼ºi¼šä¹ ä¹ ä + ...äžç” Yç%äžjæ-è • ç • CEäº K8s
äžç«i¼CEæšæœçfæf...é« - **3 ä¹ Python ä • ¥ç`äCE-
ç»éªCE**i¼šäžç š ‘ç” è,,šæœ-äºäº ‘ä¹³äºä • ¥ä... • çš,,äCEæ • è • -ä¾,,